

F - Buckets

分值: 700 分

Problem Statement

Problems F and F2 are the same problem with different constraints on M . In Problem F, $1 \leq M \leq 4$.

问题F和F2是同一道题，仅在 M 的约束条件上有所区别。在问题F中， $1 \leq M \leq 4$ 。

You are given a positive integer M . Consider the following problem.

给定一个正整数 M 。请考虑如下问题。

Bucket

You are given a positive integer N and non-negative integer sequences $A = (A_1, A_2, \dots, A_N)$ and $B = (B_1, B_2, \dots, B_N)$ of length N . All elements of A and B are between 0 and M , inclusive.

There are N buckets numbered 1 to N . Each bucket can hold up to M liters of water. Initially, bucket i contains A_i liters of water.

You may perform the following operation any number of times.

- Choose two distinct buckets i and j . Continue pouring water from bucket i into bucket j as long as both of the following conditions are satisfied:
 - Bucket i still has water remaining.
 - The amount of water in bucket j is less than M liters.

Your goal is to have exactly B_i liters of water in bucket i for all i . Determine whether the goal can be achieved.

桶

给定正整数 N ，以及两个长度为 N 的非负整数序列 $A = (A_1, A_2, \dots, A_N)$ 和 $B = (B_1, B_2, \dots, B_N)$ 。 A 和 B 的所有元素均在0到 M 之间（包含两端）。

有 N 个桶，编号为1到 N 。每个桶最多可装 M 升水。初始时，第 i 个桶装有 A_i 升水。

你可以执行任意次以下操作：

- 选择两个不同的桶 i 和 j 。只要同时满足以下两个条件，就持续将水从桶 i 倒入桶 j 中：
 - 桶 i 中还有剩余的水。
 - 桶 j 中的水量小于 M 升。

你的目标是对于所有 i ，桶 i 中恰好装有 B_i 升水。请判断是否可以达成目标。

You are given a non-negative integer matrix $X = (X_{i,j}) (0 \leq i, j \leq M)$ of $(M + 1)$ rows and $(M + 1)$ columns. Process the following queries Q times.

给定一个 $(M + 1)$ 行 $(M + 1)$ 列的非负整数矩阵 $X = (X_{i,j}) (0 \leq i, j \leq M)$ 。请处理 Q 次查询。

- You are given non-negative integers i, j, Y with $0 \leq i, j \leq M$. Change $X_{i,j}$ to Y . Then, obtain non-negative integer sequences A and B by the following procedure.
 - Initialize non-negative integer sequences A and B as empty sequences.
 - For $i = 0, 1, \dots, M$ in this order:
- 给定满足 $0 \leq i, j \leq M$ 的非负整数 i, j, Y 。将 $X_{i,j}$ 修改为 Y 。然后通过以下步骤得到非负整数序列 A 和 B ：
 - 将非负整数序列 A 和 B 初始化为空序列。
 - 按顺序对 $i = 0, 1, \dots, M$ 执行：

- For $j = 0, 1, \dots, M$ in this order:

- 按顺序对 $j = 0, 1, \dots, M$ 执行：

- Do this $X_{i,j}$ times: append i to the end of A and j to the end of B .

- 重复此操作 $X_{i,j}$ 次：将 i 追加到 A 的末尾，将 j 追加到 B 的末尾。

- Solve **Bucket** for the non-negative integer sequences A and B of length $N = \sum_{i=0}^M \sum_{j=0}^M X_{i,j}$.

- 以长度为 $N = \sum_{i=0}^M \sum_{j=0}^M X_{i,j}$ 的非负整数序列 A 和 B 作为输入，求解桶问题。

Constraints

- $1 \leq M \leq 4$
- $1 \leq Q \leq 10^6$
- $0 \leq X_{i,j}, Y \leq 10^{17}$
- $0 \leq i, j \leq M$

- $\sum_{i=0}^M \sum_{j=0}^M X_{i,j} \geq 1$ at any time.
 - All input values are integers.
 - $1 \leq M \leq 4$
 - $1 \leq Q \leq 10^6$
 - $0 \leq X_{i,j}, Y \leq 10^{17}$
 - $0 \leq i, j \leq M$
 - 任意时刻满足 $\sum_{i=0}^M \sum_{j=0}^M X_{i,j} \geq 1$ 。
 - 所有输入值均为整数。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
M Q
X0,0 X0,1 ... X0,M
X1,0 X1,1 ... X1,M
⋮
XM,0 XM,1 ... XM,M
query1
query2
⋮
queryQ
```

Each query is given in the following format:

每个查询按照以下格式给出：

```
i j Y
```

Output

Output Q lines. The i -th line should contain Yes if the goal can be achieved in query _{i} , and No otherwise.

输出 Q 行。第 i 行如果第query _{i} 次查询可以达成目标则输出Yes，否则输出No。

Sample Input 1

```
3 3
0 0 0 0
0 0 2 0
1 0 0 0
0 0 0 0
0 0 0
2 3 1
2 1 1
```

Sample Output 1

```
Yes
No
Yes
```

The first query makes $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, and we obtain $A = (1, 1, 2), B = (2, 2, 0)$.

第一次查询修改为 $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, 我们得到 $A = (1, 1, 2), B = (2, 2, 0)$.

In this case, the goal can be achieved by the following sequence of operations, for example.

这种情况下，例如可以通过以下操作序列达成目标。

- Set $i = 1, j = 2$. The amount of water in each bucket changes from $(1, 1, 2)$ to $(0, 2, 2)$.
- Set $i = 3, j = 1$. The amount of water in each bucket changes from $(0, 2, 2)$ to $(2, 2, 0)$.
- 选择 $i = 1, j = 2$ 。每个桶的水量从 $(1, 1, 2)$ 变为 $(0, 2, 2)$ 。
- 选择 $i = 3, j = 1$ 。每个桶的水量从 $(0, 2, 2)$ 变为 $(2, 2, 0)$ 。

The second query makes $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, and we obtain $A = (1, 1, 2, 2), B = (2, 2, 0, 3)$.

第二次查询修改为 $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, 我们得到 $A = (1, 1, 2, 2), B = (2, 2, 0, 3)$ 。

In this case, the goal cannot be achieved no matter how operations are performed.

这种情况下, 无论如何执行操作都无法达成目标。

The third query makes $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, and we obtain $A = (1, 1, 2, 2, 2), B = (2, 2, 0, 1, 3)$.

第三次查询修改为 $X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, 我们得到 $A = (1, 1, 2, 2, 2), B = (2, 2, 0, 1, 3)$ 。

In this case, the goal can be achieved by the following sequence of operations, for example.

这种情况下, 例如可以通过以下操作序列达成目标。

- Set $i = 1, j = 2$. The amount of water in each bucket changes from $(1, 1, 2, 2, 2)$ to $(0, 2, 2, 2, 2)$.
- Set $i = 3, j = 1$. The amount of water in each bucket changes from $(0, 2, 2, 2, 2)$ to $(2, 2, 0, 2, 2)$.
- Set $i = 4, j = 5$. The amount of water in each bucket changes from $(2, 2, 0, 2, 2)$ to $(2, 2, 0, 1, 3)$.
- 选择 $i = 1, j = 2$. 每个桶的水量从 $(1, 1, 2, 2, 2)$ 变为 $(0, 2, 2, 2, 2)$ 。
- 选择 $i = 3, j = 1$. 每个桶的水量从 $(0, 2, 2, 2, 2)$ 变为 $(2, 2, 0, 2, 2)$ 。
- 选择 $i = 4, j = 5$. 每个桶的水量从 $(2, 2, 0, 2, 2)$ 变为 $(2, 2, 0, 1, 3)$ 。

Sample Input 2

4 10

45636788580181785 16131322312654301 43477244591521823 6505049084010674 8653062732709
95921187347997793 55491565467039163 87684565747362311 80318628430974482 123080928783
75570615154690027 96403707363045776 14150012766408204 6612197700307407 6441702269296
5530468643826479 41731276604630756 15675296751519388 59461896803210859 666666666666
72767956047192820 18258893791516726 58852629621892634 3333333333333333 299239854085
2 1 26541245644686826
2 4 29485791833729050
4 1 21832826336874318
3 2 4953499115446612
2 0 69217973349997921
2 4 23133150029036944
3 4 55834224798559242
1 2 98517007615469735
2 3 3768060024403703
0 4 87241661746072372

Sample Output 2

No
Yes
No
Yes
No
Yes
No
No
No
No